

Electric Potential

Goal.

- See electric potential as charge's version of *height*: PE per charge, not PE itself.
- Ask whether V stays constant across a region where $\vec{E}$ is constant (a parallel-plate capacitor) — and check that against the same question for gravity.
- Convert between a potential difference and a change in speed (Example 25.6).
- Build V for one point charge, then for several by *adding numbers*, not vectors (Example 25.9).

1. Electric potential is charge's “height”

A ball's gravitational PE isn't just a property of the ball — you need two more things:

- _____
- Height *relative to where*? _____

Electric potential energy works the same way. You need:

- _____
- Potential *relative to where*? _____

Electric potential is potential energy *per charge* — the part of that pair that doesn't depend on which charge you put there:

$V \equiv$

measured in _____, where $1 \text{ V} =$ _____.

Check. You're told $V = 50 \text{ V}$ at some point. Can you find the PE a charge would have there? What's missing?

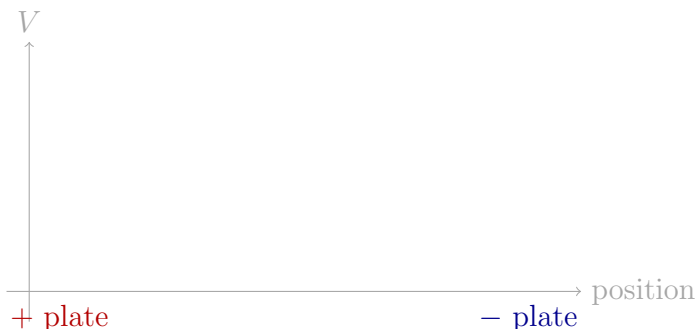
2. Does V stay constant where $\vec{E}$ is constant?

A parallel-plate capacitor: two sheets of charge, surface density $\pm\eta$, close together. Draw the field between and outside the plates.

- Each plate alone gives a field of magnitude _____ on either side of it.
- Between the plates, the two fields point the _____ direction and add: $E_{\text{total}} =$ _____ — uniform, pointing from + to −. Outside the plates they _____: $E = 0$.

Check. $\vec{E}$ is exactly constant everywhere between the plates. Does that mean V is constant there too? Compare to gravity: is gravitational PE constant wherever the *force* of gravity is constant?

Sketch V as you walk from the + plate to the − plate:



- The _____ plate is at the higher potential.
- $\vec{E}$ points from the + plate to the − plate — i.e. toward _____ V . (Same rule as always: $\vec{E}$ points “downhill.”)

3. Worked example: crossing a potential difference

A proton with speed 2.0×10^5 m/s enters a region where it crosses a potential difference of 100 V. What's its final speed? What if it were an electron instead?

Energy conservation, with $U = qV$:

$$\frac{1}{2}mv_f^2 = \frac{1}{2}mv_i^2 - q \Delta V \quad \Longrightarrow \quad v_f =$$

Proton ($q = +e$, $m = 1.67 \times 10^{-27}$ kg):

$$(v_f)_p =$$

Electron ($q = -e$, much smaller m):

$$(v_f)_e =$$

Check. The proton slowed down; the electron sped up. Both crossed the *same* potential difference. Why the opposite result?

4. Potential of a single point charge

Start from the PE of two point charges, and divide out the second charge — the same move that turned U into V up in §1.

$$U_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}} \quad \Rightarrow \quad V_1 = \frac{U_{12}}{q_2} =$$

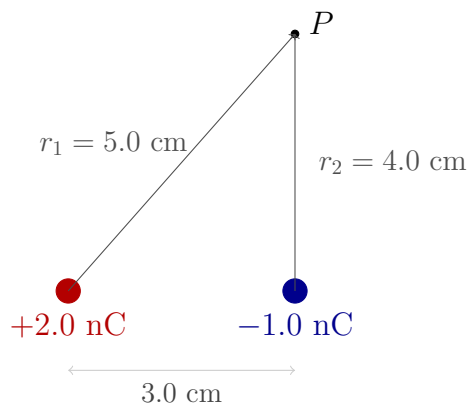
Given the potential of a point charge, what do you still need to get an actual PE out of it?

- _____
- _____ — a point charge's own potential already comes with a reference built in: $V \rightarrow 0$ as $r \rightarrow \infty$.

Check. As $r \rightarrow \infty$, what does V_1 approach? Does the *sign* of q_1 matter to V_1 ?

5. Potential from several charges: just add the numbers

Example 25.9. A $+2.0 \text{ nC}$ charge is 5.0 cm from point P ; a -1.0 nC charge is 4.0 cm from P . Find V at P .



The potential is a *scalar* — no angles, no components. Just add the two numbers:

$$V = V_1 + V_2 = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1} + \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2}$$

$V_1 =$

$V_2 =$

$V =$

Check. Suppose you needed $\vec{E}$ at P instead of V . Would this same “just add the numbers” shortcut work?